

Arnav Balodi

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Professional Summary

Aspiring coder with a strong interest in AI/ML, driven by curiosity and a commitment to continuous learning. My experience as a football player has instilled in me resilience, teamwork, and discipline, while also shaping my leadership qualities and effective communication skills. I am adaptable, quick to learn, and approach challenges with a problem-solving mindset, as well as the ability to perform under pressure.

Education

Sardar Vallabhbhai National Institute of Technology (SVNIT) <i>Bachelor of Technology in Computer Science; CGPA: 7.65</i>	Gujarat, India 2024 – 2028
DPS Modern Indian School <i>High School Diploma; 12th: 88%, 10th: 95%</i>	Doha, Qatar 2020 – 2024

Key Courses

Computer Science and Engineering: Data Structures & Algorithms, Computer Organization and Architectures (COA), Object Oriented Programming (OOPs) in Python, Database Management Systems
AI: Automata Theory, Artificial Intelligence and Techniques
Mathematics: Linear Algebra and Statistics, Discrete Math, Engineering Math, Abstract Algebra

Research Experience

Undergraduate Researcher <i>Guide: Dr. Rahul Shrivastava, NIT Surat</i>	Surat, India Present
- Conducting guided research in Recommender Systems, investigating advanced collaborative and content-based filtering methodologies to enhance prediction accuracy. - Developing a mathematical framework focused on optimizing Peak Signal-to-Noise Ratio (PSNR) and Weighted Roulette through a custom algorithmic formula to refine recommendation system performance. - Reviewing scholarly literature to analyze and implement state-of-the-art optimization techniques for large-scale data retrieval. - Formulated the Diversity-Oriented Personalized Multiplier (DOPM), a novel mathematical objective function designed to evaluate and enhance prediction accuracy in Multi-Stakeholder Recommender Systems (MSRS).	

Technical Skills

Languages: C++, HTML/CSS, MySQL, Python
Frontend & Backend: Node.js, Vite
AI/ML & Data: Reinforcement Learning, Computer Vision, Neural Networks, Pandas, Numpy
Hardware & IoT: ESP32, Arduino, Microprocessors, Microcontroller Integration
DevOps & Tools: Git/GitHub, Vercel, Linux

Engineering Projects & Experience

IT Internship – Doha Oasis <i>WSL2, Yarn, Linux, Server Infrastructure, Web Development</i>	June 2026 – July 2026
- Utilized WSL2 and Yarn to configure and streamline a Linux-native environment for local web development. - Gained hands-on knowledge in data center architecture, server initialization protocols, and network infrastructure. - Contributed to front-end updates for the company website, pitching UI/UX and functional improvements that were recognized and implemented.	
Real-Time Attendance Tracker <i>PyTorch, InsightFace, Python</i>	Mar. 2026
- Architected a sophisticated facial recognition pipeline utilizing InsightFace for high-precision identification. - Optimized model inference with PyTorch, successfully achieving a 95%+ accuracy rate across varying environmental conditions and diverse datasets.	
Search.exe Search Engine <i>JavaScript, DOM Manipulation, Document Parsing</i>	Mar. 2026
- Developed a performant, classic-style search interface capable of indexing and querying PDF and Word documents. - Engineered a custom "Dino Game" easter egg within the UI to demonstrate advanced event handling and DOM control.	
Rabuste Cafe E-Commerce & Chatbot Platform <i>React, Node.js, Vercel</i>	Jan. 2026

- Deployed a full-stack platform to digitize local cafe operations, streamlining order flow and customer management.
- Integrated an Express-based server-side chatbot to automate customer support, significantly reducing manual response latency.

Solar-Powered Water Vortex Extractor | *Arduino, ESP32 CAM*

Dec. 2025

- Designed a sustainable IoT solution utilizing motor-driven hydro-vortex technology for aquatic waste collection.
- Configured an ESP32 CAM for remote monitoring, ensuring reliable hardware-software synergy in eco-friendly infrastructure.

Certifications & Achievements

- **Smart India Hackathon (SIH) Qualifier:** Successfully cleared the internal University-level round of the national hackathon for a proposed AI-driven civic reporting solution (Civic Beacon).
- **AI Fluency for Students:** Credentialed by Anthropic Academy (2025) for proficiency in LLM application and AI ethics.
- **TechnoVision No-Code Hackathon:** Participated in WPU CSI competition (Nov 2025), building rapid prototype solutions for real-world problems.
- **Competitive Programming:** Actively solving complex algorithmic challenges on LeetCode and Codeforces.
- **Google Winter of Code (GWOC):** Participant; contributed to open-source projects, enhancing technical collaboration and code optimization skills.
- **CISCO PCAP:** Python programming essentials certificate accredited by CISCO.
- **Udemy Master C++:** Professional Programming Certificate accredited by Udemy Academy.
- **Google Cloud Study Jams:** Completed comprehensive hands-on training via the Google Cloud console, earning 19 Skill Badges in domains including Generative AI, Vertex AI, Infrastructure Modernization, and Data Management.